EIA CASE STUDY

Kochi Waste-to-Energy Project Brahmapuram, Kerala

Solid Waste Management Project

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Introduction

Why Brahmapuram became an important solid-waste case study

The problem

Kochi's waste accumulated at Brahmapuram because existing processing capacity and management were inadequate. Uncontrolled dumping created odour, leachate, fire and pollution risks.

The proposed response

A Waste-to-Energy (WtE) facility was proposed to process municipal solid waste, recover energy and reduce dependence on open dumping.

Why EIA matters

The project could reduce waste-related impacts, but combustion, wastewater, traffic, ash and emissions also required environmental assessment.

- Project proponent: GJ Eco Power Private Limited, with Kochi Municipal Corporation as the local authority.
- EIA/EMP prepared by L&T Infrastructure Engineering Ltd. in 2019.
- The project later became a useful case study because its approval, public concerns and eventual termination show why environmental assessment is important.

Location: The Brahmapuram Waste-to-Energy project site is located in the **Vadavucode-Puthencruz panchayat** area on the **southeastern outskirts of Kochi (Cochin)**, **Ernakulam District, Kerala, India**

Purpose and Objectives

What the project was intended to achieve

1	**Manage municipal solid waste**	Process incoming MSW instead of continued uncontrolled dumping.
2	**Recover energy**	Convert suitable waste-derived fuel into electricity.
3	**Reduce landfill dependence**	Reduce the volume of waste requiring final disposal.
4	**Address legacy waste**	Use the project as part of the wider strategy for the accumulated Brahmapuram waste.
5	**Improve environmental quality**	Reduce odour, fire risk, leachate and waste-related pollution.

Project Description

•Waste collected from Kochi and nearby areas would be brought to the facility for weighing, sorting and processing.

•Recyclable and recoverable materials would be separated from the incoming waste.

•The remaining combustible waste would be processed into Refuse-Derived Fuel (RDF).

•The prepared waste/RDF would be fed into a gasification system, where it is converted into combustible gas.

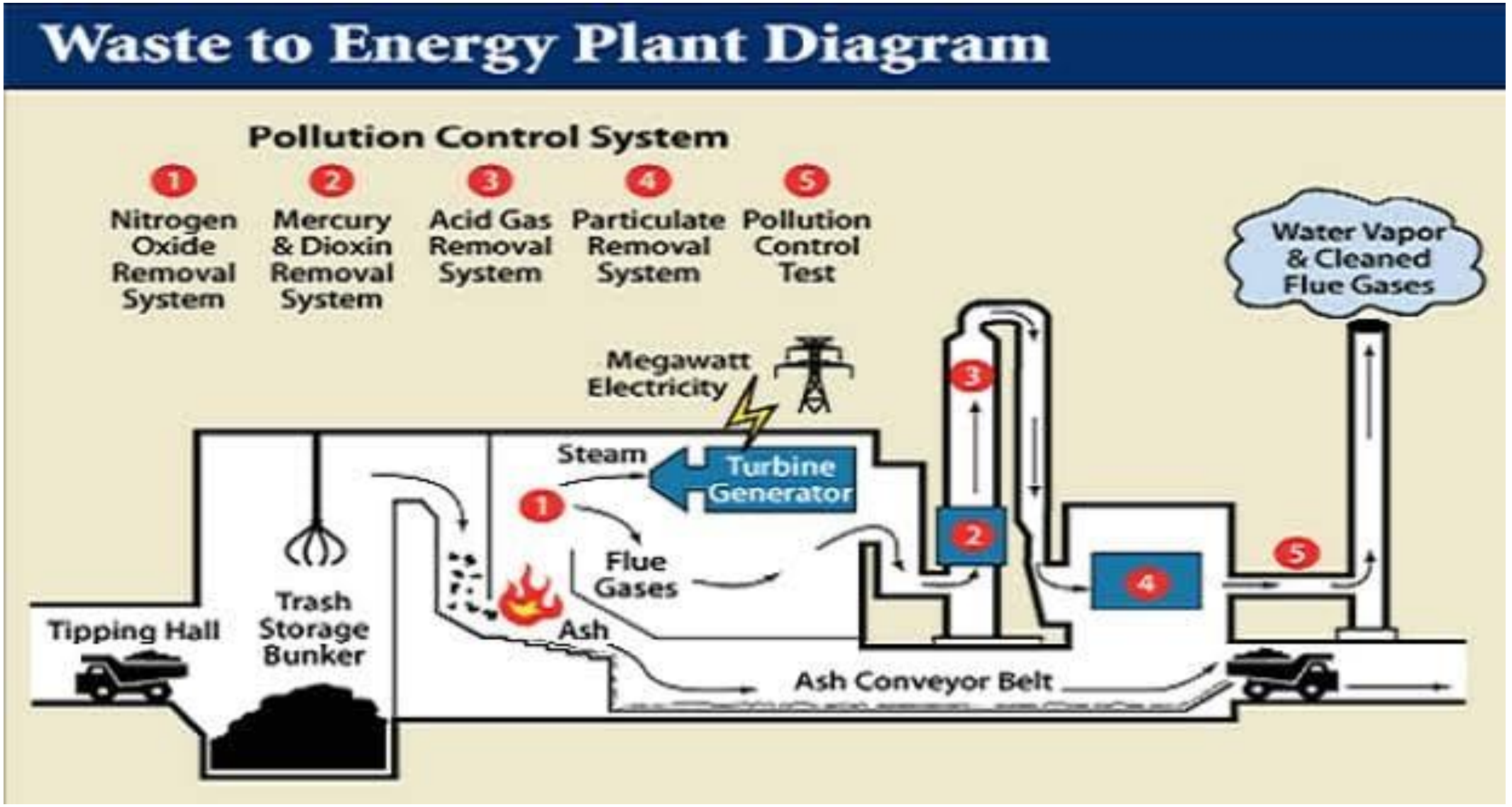
•This gas would be used to generate electricity, with a proposed installed capacity of about 12.65 MW.

•The process would also produce ash and other residues, which would be collected and safely managed.

•The facility would include systems for waste handling, bio-drying, segregation, gasification, power generation and pollution-control measures.

•The main purpose is to reduce dependence on dumping at the Brahmapuram waste site while recovering energy from municipal waste.

Working


Waste to Energy Plant Diagram
Pollution Control System
1
Nitrogen Oxide Removal System
2
Mercury & Dioxin Removal System
3
Acid Gas Removal System
4
Particulate Removal System
5
Pollution Control Test
Water Vapor & Cleaned Flue Gases
Megawatt Electricity
Steam
Turbine Generator
Flue Gases
Ash
Tipping Hall
Trash Storage Bunker
Ash Conveyor Belt

Why Was an EIA Required?

Potential benefits and risks had to be assessed together

Air emissions

Gasifier/boiler stack, emergency DG set and vehicle movement could affect air quality.

Water & leachate

Waste reception and bio-drying can generate leachate; the site is close to water bodies.

Soil & land

Long-term waste handling can cause contamination and land-use impacts.

Noise & traffic

Construction, plant equipment and hundreds of waste-transport movements can create noise and congestion.

Ecology

The low-lying site and nearby water environment required assessment of ecological sensitivity.

Community health

Odour, fire, smoke, waste transport and perceived health risks were major local concerns.

Baseline Environmental Conditions

What the EIA studied before project implementation

Water environment (PH:4.88-6.69)

Surface water and groundwater were included in baseline surveys. Site is low-lying and adjacent to the Kadambrayar River/canal system; monsoon inundation was considered.

Air environment (PM 10: 33.6)

Ambient air monitoring was carried out at multiple locations. The EIA used site-specific meteorological data and air-dispersion modelling.

Noise environment

Ambient noise was part of the baseline assessment around the project influence area.

Soil / land

Soil quality and land conditions were surveyed; the existing waste facility was already an important source of environmental pressure.

Ecology

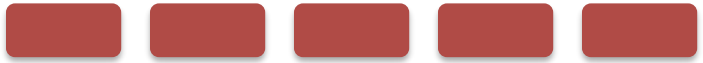
Flora and fauna surveys were carried out. Vembanad Lake and other sensitive receptors were considered in the study area.

Socio-economic

The project site had no existing settlements requiring R&R. The EIA identified potential employment and local infrastructure benefits.

Baseline Snapshot

Important observations from the EIA study area

Existing waste pressure		Major existing issue
Water sensitivity		Needs protection
Air/odour concern		Major existing issue
Ecological sensitivity		Requires care
Community sensitivity		Strong public interest

Anticipated Environmental Impacts – Construction

Air & dust

Earthwork, material handling, construction vehicles and DG sets may increase dust and particulate emissions.

Noise & vibration

Construction machinery and truck movement can affect nearby receptors.

Water & soil

Runoff, accidental spills and poor waste storage could contaminate soil or nearby water.

Traffic & safety

Movement of heavy vehicles increases road-use pressure and construction safety requirements.

Employment & local economy

Construction can generate temporary jobs and demand for local services — a positive socio-economic effect.

• Overall character: mostly temporary and manageable if good construction practices, dust suppression, safe material storage and traffic management are enforced.

Environmental Impacts – Operation

Positive impacts

- Scientific processing of MSW
- Reduced uncontrolled dumping
- Lower landfill dependence
- Electricity generation from waste-derived fuel
- Potential reduction in odour and fire risk at the old dump
- Employment and skill development
- Potential recovery of land from legacy waste

Negative / residual impacts

- Flue-gas emissions from thermal processing
- Leachate and wastewater generation
- Ash and reject disposal
- Noise and truck traffic
- Odour if waste handling is poorly controlled
- Risk of contamination if treatment systems fail
- Community concerns about health and emissions

Summary of Identified Environmental impacts

Environmental Component	Construction Measures	Operation Measures	Expected Outcome
Air Quality	Water sprinkling, covered trucks, dust control	Stack emission control, regular vehicle maintenance, odour control	Reduced dust, emissions & odour
Water Resources	Proper drainage, prevent runoff from construction area	Leachate collection & treatment, impermeable flooring, ETP	Prevent groundwater & surface-water contamination
Noise Environment	Low-noise equipment, restricted working hours	Equipment maintenance, acoustic enclosures	Noise kept within permissible limits
Soil & Land	Minimise vegetation clearing, proper waste disposal	Lined waste-handling areas, controlled ash/residue disposal	Prevent soil contamination
Ecology	Minimise disturbance to surrounding vegetation	Greenbelt development, controlled emissions/discharges	Reduced ecological disturbance
Socio-Economic	Local employment opportunities, traffic management	Improved waste management, electricity generation, local employment	Long-term community benefit

Environmental Management Measures

Key controls proposed in the EIA/EMP

Air pollution

Baghouse filter; lime-based acid-gas removal; De-NOx/flue-gas recirculation; SNCR; continuous flue-gas monitoring.

Wastewater

Leachate collection; underground storage; reed-bed/MSR treatment; UF/RO treatment where required.

Leachate reuse

Collected leachate from bio-drying areas is managed within the process to reduce uncontrolled discharge.

Ash & rejects

Segregation and controlled disposal of residual ash/inert material; landfill only for permitted residuals.

Odour & hygiene

Enclosed/controlled waste handling, rapid processing and good housekeeping to reduce nuisance.

Greenbelt & safety

Greenbelt development, fire protection, worker PPE and emergency-response provisions.

Environmental Monitoring

Monitoring is essential to verify that mitigation works

Parameter	Example monitoring focus	Frequency / approach
Air emissions	Stack gases, dust and combustion-related pollutants	Continuous/periodic as specified; compliance with standards
Ambient air	PM and gaseous pollutants at receptor locations	Periodic monitoring
Groundwater	Water level and quality near waste/landfill areas	Groundwater level annually; quality as specified
Leachate	pH, BOD, COD, TSS/TDS, nutrients and metals	Once every 3 months
Noise	Ambient noise around sensitive receptors	Periodic compliance monitoring
Soil	Physico-chemical quality and heavy metals	Periodic monitoring
Methane / hydrocarbons	Fugitive gases near relevant plant areas	Daily gas-detection checks

Public Consultation

Environmental concerns were a major part of the project debate

Major concerns raised

- Existing Brahmapuram pollution was already severe.
- Fear of toxic emissions from waste combustion.
- Concerns about water, air and health impacts.
- Questions about waste availability, segregation and technology suitability.
- Demand for stronger source-level waste treatment.

How the EIA addressed them

- Baseline environmental surveys and receptor mapping.
- Air-dispersion modelling.
- Flue-gas treatment and continuous monitoring provisions.
- Wastewater/leachate treatment systems.
- EMP, environmental monitoring and emergency measures.

Public hearing

A formal environmental public hearing was notified for 10 June 2019. The project was to be appraised by SEIAA Kerala after the EIA and public consultation process.

Environmental Clearance & Project Status

The case illustrates that clearance is not the same as successful implementation



EIA + public hearing
SEIAA appraisal process

SEIAA records that EC
had been given to GJ Eco Power

GJ Eco Power project
later terminated

Kerala Govt. selected a
new 300 TPD WtE concessionaire

Brahmapuram's waste system
includes a commissioned 150 TPD CBG plant

Major Findings

What the case study tells us

1. Environmental need was urgent because uncontrolled waste disposal was already causing air, water, soil, odour and fire-related problems

2. The proposed WtE project offered significant potential benefits through waste diversion and energy recovery.

3. Major residual risks were associated with emissions, wastewater/leachate, ash/rejects, traffic and community acceptance.

4. Baseline surveys, modelling and an EMP were essential because the site was environmentally sensitive and already degraded.

5. Public consultation highlighted the importance of source segregation, local health concerns and confidence in pollution controls.

6. The later termination of the original project demonstrates that environmental clearance alone does not guarantee successful implementa

Overall Environmental Significance

A balanced EIA perspective

Environmental need

HIGH

The project was proposed in response to a serious waste-management problem and legacy waste accumulation.

The “do nothing” scenario also carries environmental costs.

Potential benefit

HIGH

Waste diversion + resource recovery + electricity generation + reduced open dumping can improve environmental performance if controls work.

Risk management

CRITICAL

Strict emission control, wastewater treatment, monitoring, source segregation, safe ash handling and public transparency are necessary.

Conclusion

Key takeaway

Conclusion

The Brahmapuram case shows why EIA is essential in solid waste management. A waste-processing project can provide major environmental benefits, but only when technology, pollution controls, monitoring, community concerns and long-term waste strategy are addressed together.

What a good project needs

- Source segregation
- Reliable treatment technology
- Strong pollution controls
- Continuous monitoring
- Safe residual disposal
- Community participation

Final lesson

EIA should not be treated as a paperwork requirement. It is a decision-making tool that helps compare environmental costs, benefits, alternatives and safeguards before and during a project.

References

Primary and supporting sources used for this presentation

- L&T Infrastructure Engineering Ltd. (2019). Draft EIA/EMP Report – EC for Kochi Waste to Energy Project, GJ Eco Power Pvt. Ltd.
- State Environment Impact Assessment Authority (SEIAA), Kerala – meeting minutes on the proposed Kochi Waste to Energy Project at Brahmapuram.
- Government of Kerala, Local Self Government Department – orders on integrated solid waste management / waste-to-energy projects at Brahmapuram.
- Kochi Municipal Corporation / Kerala Government – reports on the Brahmapuram solid waste management facility.
- National Green Tribunal records and Kochi Municipal Corporation action-taken reports on solid waste management at Brahmapuram.
- Press Information Bureau, Government of India (27 Feb 2026) – inauguration of BPCL's 150 TPD CBG plant at Brahmapuram.
- Supporting research/news sources: Times of India; Kerala State Electricity Regulatory Commission records; peer-reviewed/academic RIAM study on Brahmapuram.

Note: PPT uses concise paraphrased points; detailed explanations should be given during the presentation.